

Tangent and Normal Spaces

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Consider the map (in general nonlinear)

$$\mathbf{g} : \mathbb{R}^n \longrightarrow \mathbb{R}^m \quad (1)$$

given by

$$\mathbf{g}(\mathbf{A}) = \begin{pmatrix} g_1(\mathbf{A}) \\ \vdots \\ g_m(\mathbf{A}) \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \vdots \\ \rho_m \end{pmatrix} = \boldsymbol{\rho} \in \mathbb{R}^m; \mathbf{A} \in \mathbb{R}^n,$$

that defines a manifold

$$\mathfrak{M} = \{ \mathbf{A} \mid \mathbf{g}(\mathbf{A}) = \boldsymbol{\rho} \}, \quad (2)$$

which is a level surface in $\mathbb{R}^{n+m}$.

As a special case one can consider the bilinear function

$$\mathbf{f}(A) = \left(A \mid \widehat{\Phi^\dagger \Phi_\varphi} A \right), \quad A \in \mathcal{B}_{sa}(\mathcal{H}) \cong \mathbb{R}^{n^2}; n = \binom{m}{N}; \quad \mathbf{f}(A) \in \mathbb{R}^m \quad (3)$$

where $\Phi^\dagger \Phi_\varphi$ is a fixed vector operator with components

$$\Phi^\dagger \Phi_{\varphi_j} = \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \quad (4)$$

formed by the comb $\{\varphi_j \mid 1 \leq j \leq m\}$ for $\mathcal{H}$ and

$$\widehat{\Phi^\dagger \Phi_{\varphi_j}} |A\rangle = \frac{1}{2} \left[\Phi(\varphi_j)^\dagger \Phi(\varphi_j), A \right]_+ \equiv \widehat{\Phi^\dagger \Phi_{\varphi_j}} A = \frac{1}{2} \left[\Phi(\varphi_j)^\dagger \Phi(\varphi_j), A \right]_+. \quad (5)$$

As $m \rightarrow \infty$ the operators $\Phi(\varphi_j)^\dagger \Phi(\varphi_j) \rightarrow \Phi(\delta_{\mathbf{r}})^\dagger \Phi(\delta_{\mathbf{r}})$ and $\Phi^\dagger \Phi_\varphi \rightarrow \Phi^\dagger \Phi_\delta$ i.e. the vector operator tends to a distribution operator

$$\Phi^\dagger \Phi_\varphi : \mathbb{N} \rightarrow \mathcal{B}(\mathcal{H}) \quad (6)$$

$$\Phi^\dagger \Phi_\delta : \mathcal{D}_\delta \equiv \mathbb{R}^3 \rightarrow \mathcal{B}(\mathcal{H}) \quad (7)$$

and $\Phi^\dagger \Phi_\varphi$ defines the super operator $\widehat{\Phi^\dagger \Phi_\varphi}$ by symmetrized multiplication. Noting that

$$A = \sum_{1 \leq p, q \leq n} A_{pq} |\Psi_p\rangle \langle \Psi_q| \quad (8)$$

this quadratic function can be written in detail as

$$\mathbf{f}_j(A) = \left(A \left| \widehat{\Phi^\dagger \Phi}_{\varphi_j} A \right. \right) = \frac{1}{2} \text{Tr} \left\{ A \left[\Phi(\varphi_j)^\dagger \Phi(\varphi_j), A \right]_+ \right\} \quad (9)$$

$$= \frac{1}{2} \text{Tr} \left\{ A \left(\Phi(\varphi_j)^\dagger \Phi(\varphi_j) A + A \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \right) \right\} = \text{Tr} \left\{ \Phi(\varphi_j)^\dagger \Phi(\varphi_j) A^2 \right\} \quad (10)$$

$$= \sum_{1 \leq p, q, s \leq n} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle A_{pq} A_{qs} \quad (11)$$

The derivatives of $\mathbf{f}$ at the point A are then

$$\frac{\partial \mathbf{f}_j(A)}{\partial A_{\alpha\beta}} = \sum_{1 \leq p, q, s \leq n} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle \{ \delta_{\alpha p} \delta_{\beta q} A_{qs} + A_{pq} \delta_{\alpha q} \delta_{\beta s} \} \quad (12)$$

$$= \sum_{1 \leq s \leq n} \left\{ A_{\beta s} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_\alpha \right. \right\rangle + \left\langle \Psi_\beta \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_s \right. \right\rangle A_{s\alpha} \right\} \quad (13)$$

$$= \sum_{1 \leq s \leq n} \left\{ A_{\beta s} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_\alpha \right. \right\rangle + \left\langle \Psi_\beta \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_s \right. \right\rangle A_{s\alpha} \right\} \quad (14)$$

$$\frac{\partial^2 \mathbf{f}_j(A)}{\partial A_{\lambda\mu} \partial A_{\alpha\beta}} = \left\langle \Psi_\mu \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_\alpha \right. \right\rangle \delta_{\lambda\beta} + \left\langle \Psi_\beta \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_\lambda \right. \right\rangle \delta_{\mu\alpha} \quad (15)$$

For

$$A \rightarrow A + \delta A \quad (16)$$

(if δA belongs to the tangent space then $A + \delta A$ does not belong to the surface) i.e. expanding for each component in a small nbhd of A we get

$$\mathbf{f}_j(A + \delta A) = \sum_{1 \leq p, q, s \leq n} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle (A + \delta A)_{pq} (A + \delta A)_{qs} \quad (17)$$

$$= \sum_{1 \leq p, q, s \leq n} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle A_{pq} A_{qs} \quad (18)$$

$$+ \sum_{1 \leq p, q, s \leq n} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle \{ \delta A_{pq} A_{qs} + A_{pq} \delta A_{qs} \} \quad (19)$$

$$+ \sum_{1 \leq p, q, s \leq n} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle \delta A_{pq} \delta A_{qs} \quad (20)$$

giving

$$= \rho_j + \sum_{1 \leq p, q, s \leq n} \left\{ A_{qs} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle \delta A_{pq} + \delta A_{qs} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle A_{pq} \right\} \quad (21)$$

$$+ \sum_{1 \leq p, q, s \leq n} \delta A_{qs} \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle \delta A_{pq} \quad (22)$$

(the last term can be written as $\sum_{1 \leq p, s \leq n} \langle \Psi_s | \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \rangle (\delta A)_{ps}^2$) producing

$$\mathbf{f}(A) = \boldsymbol{\rho} + 2 \begin{pmatrix} \vdots & \dots & \vdots \\ j \uparrow \vdots & \langle \Psi_q | A \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \rangle & \vdots \\ \vdots & \dots & \vdots \end{pmatrix} \begin{pmatrix} \vdots \\ \delta A_{pq} \\ \vdots \end{pmatrix} \quad (23)$$

$$+ \frac{1}{2} \begin{pmatrix} \dots & \delta A_{pq} & \dots \end{pmatrix} \begin{pmatrix} \vdots & \dots & \vdots \\ \vdots & \langle \Psi_s | \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \rangle \delta_{rq} & \vdots \\ \vdots & \dots & \vdots \end{pmatrix} \begin{pmatrix} \vdots \\ \delta A_{rs} \\ \vdots \end{pmatrix} \quad (24)$$

Assuming the function $\mathbf{g}(\mathbf{A})$ is Fréchet differentiable

$$\begin{aligned} [d\mathbf{g}(\mathbf{A})](\delta\mathbf{A}) &= \begin{pmatrix} \frac{\partial g_1(\mathbf{A})}{\partial A_1} & \dots & \frac{\partial g_1(\mathbf{A})}{\partial A_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m(\mathbf{A})}{\partial A_1} & \dots & \frac{\partial g_m(\mathbf{A})}{\partial A_n} \end{pmatrix} \begin{pmatrix} \delta A_1 \\ \vdots \\ \delta A_n \end{pmatrix} = \mathbb{D}\mathbf{g}(\mathbf{A})|\delta\mathbf{A} \\ &= \begin{pmatrix} \nabla g_1(\mathbf{A}) \\ \vdots \\ \nabla g_m(\mathbf{A}) \end{pmatrix} \begin{pmatrix} \delta A_1 \\ \vdots \\ \delta A_n \end{pmatrix} = \begin{pmatrix} \delta \rho_1 \\ \vdots \\ \delta \rho_m \end{pmatrix} \in \mathbb{R}^m. \end{aligned}$$

$$\begin{aligned} [d^2\mathbf{g}(\mathbf{A})](\delta\mathbf{A}) &= \frac{1}{2} \begin{pmatrix} \delta A_1 & \dots & \delta A_n \end{pmatrix} \begin{pmatrix} \nabla^2 g_1(\mathbf{A}) \\ \vdots \\ \nabla^2 g_m(\mathbf{A}) \end{pmatrix} \begin{pmatrix} \delta A_1 \\ \vdots \\ \delta A_n \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \begin{pmatrix} \delta A_1 & \dots & \delta A_n \end{pmatrix} \nabla^2 g_1(\mathbf{A}) \begin{pmatrix} \delta A_1 \\ \vdots \\ \delta A_n \end{pmatrix} \\ \vdots \\ \begin{pmatrix} \delta A_1 & \dots & \delta A_n \end{pmatrix} \nabla^2 g_m(\mathbf{A}) \begin{pmatrix} \delta A_1 \\ \vdots \\ \delta A_n \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \delta^2 \rho_1 \\ \vdots \\ \delta^2 \rho_m \end{pmatrix} \in \mathbb{R}^m. \end{aligned}$$

For the above example

$$\frac{\partial f_j(A)}{\partial A_{pq}} = \left(A | \Phi(\varphi_j)^\dagger \Phi(\varphi_j) | \varphi_p \rangle \langle \varphi_q | \right) = \text{Tr} \left\{ A \Phi(\varphi_j)^\dagger \Phi(\varphi_j) | \varphi_p \rangle \langle \varphi_q | \right\} \quad (25)$$

$$= \langle \varphi_q | A \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \varphi_p \rangle; \quad 1 \leq j, p, q \leq m; n = m^2 \quad (26)$$

and

$$d\mathbf{f}(A) = \left(\widehat{\Phi^\dagger \Phi_\varphi} A \right) = \frac{1}{2} \left([\Phi^\dagger \Phi_\varphi, A]_+ \right). \quad (27)$$

The second derrivatives are

$$(\nabla^2 f_j(A))_{pqrs} = \left\langle \Psi_s \left| \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \Psi_p \right. \right\rangle \delta_{rq}; \quad 1 \leq p, q, r, s \leq m. \quad (28)$$

The kernel, $\ker \{d\mathbf{g}(\mathbf{A})\}$, of $d\mathbf{g}(\mathbf{A})$ is defined as

$$\ker \{d\mathbf{g}(\mathbf{A})\} = \{\delta \mathbf{A} \mid [d\mathbf{g}(\mathbf{A})](\delta \mathbf{A}) = \mathbf{0}\} \subseteq \mathbb{R}^n.$$

The tangent space $\mathcal{T}_{\mathbf{A}}$ of $\mathbf{g}$ at the point $\mathbf{A}$ is contained in this kernel. A tangent vector $\delta \mathbf{A}$ at the point $\mathbf{A}$ is defined by considering curves $\in \mathfrak{M}$

$$\mathbf{g}(\mathbf{x}(t)) = \rho, t \in [-\delta, \delta]$$

$$\mathbf{x}(0) = \mathbf{A}$$

$$\mathbf{x} : [-\delta, \delta] \longrightarrow \mathbb{R}^n$$

and is defined to be formed by the equivalence class of velocity vectors to such curves at the point $\mathbf{A}$ i.e. $\left\{ \frac{d\mathbf{x}(0)}{dt} \right\}$. The Taylor series of $\mathbf{x}$ around 0 is

$$\mathbf{x}(t) = \mathbf{A} + \frac{d\mathbf{x}(0)}{dt}t + \frac{1}{2} \frac{d^2\mathbf{x}(0)}{dt^2}t^2 + \dots \quad (29)$$

and of $\mathbf{g}$ restricted to the curve $\mathbf{x}$ that lies in the surface $\mathfrak{M}$ is

$$\mathbf{g}(\mathbf{x}(t)) = \rho + \frac{d\mathbf{g}(\mathbf{x}(0))}{dt}t + \frac{1}{2} \frac{d^2\mathbf{g}(\mathbf{x}(0))}{dt^2}t^2 + \dots = \rho \quad (30)$$

where

$$\frac{d\mathbf{g}(\mathbf{x}(0))}{dt} = \mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \frac{d\mathbf{x}(0)}{dt} \quad (31)$$

$$\frac{d^2\mathbf{g}(\mathbf{x}(0))}{dt^2} = \left\{ \frac{d}{dt} \mathbb{D}_x \mathbf{g}(\mathbf{x}(t)) \right\} \Big|_{t=0} \frac{d\mathbf{x}(0)}{dt} + \mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \frac{d^2\mathbf{x}(0)}{dt^2} \quad (32)$$

$$= \left(\frac{d\mathbf{x}(0)}{dt} \right)^t \mathbb{D}_x^2 \mathbf{g}(\mathbf{x}(0)) \frac{d\mathbf{x}(0)}{dt} + \mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \frac{d^2\mathbf{x}(0)}{dt^2}. \quad (33)$$

Hence

$$\frac{d\mathbf{g}(\mathbf{x}(t))}{dt} = \frac{d\mathbf{g}(\mathbf{x}(0))}{dt} + \frac{d^2\mathbf{g}(\mathbf{x}(0))}{dt^2}t + \dots = \mathbf{0} \quad (34)$$

and *in particular* at $t = 0$

$$\frac{d\mathbf{g}(\mathbf{x}(0))}{dt} = \mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \frac{d\mathbf{x}(0)}{dt} = \mathbf{0}, \quad (35)$$

i.e.

$$\frac{d\mathbf{x}(0)}{dt} \in \ker \{ \mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \}. \quad (36)$$

The derivatives wrt x are the same as wrt $\mathbf{A}$ thus

$$\mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) = \mathbb{D} \mathbf{g}(\mathbf{A}) \quad (37)$$

$$\mathbb{D}_x^2 \mathbf{g}(\mathbf{x}(0)) = \mathbb{D}^2 \mathbf{g}(\mathbf{A}), \quad (38)$$

thus

$$\frac{d\mathbf{x}(0)}{dt} \in \ker \{ \mathbb{D} \mathbf{g}(\mathbf{A}) \}. \quad (39)$$

One should note that

$$\delta \mathbf{A} \in \ker \{ \mathbb{D} \mathbf{g}(\mathbf{A}) \} \quad (40)$$

does not imply that $\exists \frac{d\mathbf{x}(0)}{dt} \ni \delta \mathbf{A} = \frac{d\mathbf{x}(0)}{dt}$ i.e. this condition is necessary but not sufficient for $\delta \mathbf{A}$ to be a tangent vector. One can write the above as

$$[d\mathbf{g}(\mathbf{x}(0))] (\delta \mathbf{A}) = \mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \begin{pmatrix} \frac{dx_1(0)}{dt} \\ \vdots \\ \frac{dx_n(0)}{dt} \end{pmatrix} = \mathbf{0}.$$

In detail this gives

$$\mathbb{D}_x \mathbf{g}(\mathbf{x}(0)) \begin{pmatrix} \frac{dx_1(0)}{dt} \\ \vdots \\ \frac{dx_n(0)}{dt} \end{pmatrix} = \begin{pmatrix} \nabla g_1(\mathbf{A}) \\ \vdots \\ \nabla g_m(\mathbf{A}) \end{pmatrix} \begin{pmatrix} \frac{dx_1(0)}{dt} \\ \vdots \\ \frac{dx_n(0)}{dt} \end{pmatrix} \quad (41)$$

$$= \begin{pmatrix} \frac{\partial g_1(\mathbf{A})}{\partial \mathbf{A}_1} & \dots & \frac{\partial g_1(\mathbf{A})}{\partial \mathbf{A}_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m(\mathbf{A})}{\partial \mathbf{A}_1} & \dots & \frac{\partial g_m(\mathbf{A})}{\partial \mathbf{A}_n} \end{pmatrix} \begin{pmatrix} \frac{dx_1(0)}{dt} \\ \vdots \\ \frac{dx_n(0)}{dt} \end{pmatrix} = 0. \quad (42)$$

For the example $\mathbf{f}(A)$ this gives that if $\delta A \in \mathcal{T}_A$ then

$$[d\mathbf{f}(A)](\delta A) = \left(A | \widehat{\Phi^\dagger \Phi_\varphi} \delta A \right) = \mathbf{0}, \quad (43)$$

i.e. if $\delta A \in \mathcal{T}_A$ then $|\delta A\rangle \perp \left[\Phi(\varphi_j)^\dagger \Phi(\varphi_j), A \right]_+$ $\forall j$, but being orthogonal $\forall j$ does not mean it is in the tangent space. One should note that $\left(\widehat{\Phi^\dagger \Phi_\varphi} A \right)$ is not an operator but a *vector*

operator with operator components $\left[\Phi(\varphi_j)^\dagger \Phi(\varphi_j), A \right]_+$, $1 \leq j \leq m$, so the condition eq. (43) gives m equations

$$\left(A | \widehat{\Phi(\varphi_j)^\dagger \Phi(\varphi_j)} \delta A \right) = \text{Tr} \left\{ A \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \delta A \right\} = 0; \quad 1 \leq j \leq m. \quad (44)$$

Note that if $\delta A \in \mathcal{T}_A$ then $A + \delta A$ does not belong to the surface.

The tangent space must be at least $n - m$ dimensional as

$$d\mathbf{g}(\mathbf{A}) : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad (45)$$

and the equation

$$d\mathbf{g}(\mathbf{A})(\delta \mathbf{A}) = \mathbf{0}$$

must have at least $n - m$ linearly independent solutions. It might have more solutions, which could produce a larger tangent space.

One can decompose $\mathbb{R}^n$ into the direct sum

$$\mathbb{R}^n = \ker \{d\mathbf{g}(\mathbf{A})\} \oplus \ker \{d\mathbf{g}(\mathbf{A})\}^c$$

and

$$d\mathbf{g}(\mathbf{A}) : \ker \{\mathbf{A}\mathbf{g}(\mathbf{A})\}^c \xrightarrow{1-1} \text{Range} \{d\mathbf{g}(\mathbf{A})\}.$$

The normal space of $\mathbf{g}$ at the point $\mathbf{A}$ is

$$\mathcal{N}_{\mathbf{A}}\mathbf{g} = \text{Range} \{d\mathbf{g}(\mathbf{A})\} \subseteq \mathbb{R}^m.$$

$$\mathbb{R}^m = \text{Range} \{d\mathbf{g}(\mathbf{A})\} \oplus \text{Range} \{d\mathbf{g}(\mathbf{A})\}^c$$

$$\cup \qquad \qquad \qquad \cap$$

$$\mathbb{R}^m = [\ker \{d\mathbf{g}(\mathbf{A})\}^c] \oplus [\ker \{d\mathbf{g}(\mathbf{A})\}^c]^c.$$

Observe that

$$\mathbf{g} : V \longrightarrow W \equiv \mathbf{g} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

as theses are linear spaces

$$d\mathbf{g}(\mathbf{A}) : V \longrightarrow W \equiv d\mathbf{g}(\mathbf{A}) : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

$$\langle \bullet, \bullet \rangle_V : V^d \times V \longrightarrow \mathbb{R}$$

$$\langle \bullet, \bullet \rangle_W : W^d \times W \longrightarrow \mathbb{R}$$

$$\begin{aligned}
d\mathbf{g}(\mathbf{A})^\dagger : W^d &\longrightarrow V^d \\
\left\langle d\mathbf{g}(\mathbf{A})^\dagger \mathbf{w}_d \middle| \mathbf{v} \right\rangle_V &= \langle \mathbf{w}_d | d\mathbf{g}(\mathbf{A}) \mathbf{v} \rangle_W \forall \mathbf{v} \in \text{domain} \{d\mathbf{g}(\mathbf{A})\} \subseteq V \\
\dagger : \mathcal{L}(V, W) &\longrightarrow \mathcal{L}(W^d, V^d)
\end{aligned}$$

[

$${}_V \left\langle d\mathbf{g}(\mathbf{A})^\dagger \mathbf{w}_A \right| = {}_W \langle \mathbf{w}_A | d\mathbf{g}(\mathbf{A}) = (d\mathbf{g}(\mathbf{A}) | \mathbf{w} \rangle_W)^\dagger$$

if W can be embedded in W^A]

Hence

$$\begin{aligned}
\text{Range} \{d\mathbf{g}(\mathbf{A})\}^\perp &= \{ \mathbf{w}_d \in W^d \mid \langle \mathbf{w}_d | \mathbf{w} \rangle_W = 0 \forall \mathbf{w} \in \text{Range} \{d\mathbf{g}(\mathbf{A})\} \subseteq W \} \\
&\equiv \{ \mathbf{w}_d \in W^d \mid \langle \mathbf{w}_d | \mathbf{w} \rangle_W = 0 \forall \mathbf{w} \in \text{Range} \{d\mathbf{g}(\mathbf{A})\} \subseteq \mathbb{R}^m \}
\end{aligned}$$

and

$$\begin{aligned}
\ker \{d\mathbf{g}(\mathbf{A})\}^\perp &= \{ \mathbf{v}_A \in V^d \mid \langle \mathbf{v}_d | \mathbf{v} \rangle_V = 0 \forall \mathbf{v} \in \ker \{d\mathbf{g}(\mathbf{A})\} \subseteq V \} \\
&\equiv \{ \mathbf{v}_A \in V^d \mid \langle \mathbf{v}_d | \mathbf{v} \rangle_V = 0 \forall \mathbf{v} \in \ker \{d\mathbf{g}(\mathbf{A})\} \subseteq \mathbb{R}^n \}.
\end{aligned}$$

If $\text{Range} d\mathbf{g}(\mathbf{A}) = \mathbb{R}^m$ then $\{\nabla g_j(\mathbf{A}) \mid 1 \leq j \leq m\}$ are linearly independent and the vector $\langle \mathbf{v} | \in \mathbb{R}^m$ can be uniquely expanded as

$$\begin{aligned}
\langle \mathbf{v} | &= \sum_{1 \leq j \leq m} \lambda_j \nabla g_j(\mathbf{A}) = \left(\lambda_1, \dots, \lambda_m \right) \begin{pmatrix} \nabla g_1(\mathbf{A}) \\ \vdots \\ \nabla g_m(\mathbf{A}) \end{pmatrix} \\
&= \left(\lambda_1, \dots, \lambda_m \right) \begin{pmatrix} \frac{\partial g_1(\mathbf{A})}{\partial \mathbf{A}_1} & \dots & \frac{\partial g_1(\mathbf{A})}{\partial \mathbf{A}_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m(\mathbf{A})}{\partial \mathbf{A}_1} & \dots & \frac{\partial g_m(\mathbf{A})}{\partial \mathbf{A}_n} \end{pmatrix} \\
&= \langle \boldsymbol{\lambda} | \nabla \mathbf{g}(\mathbf{A}) \equiv \langle \boldsymbol{\lambda} | d\mathbf{g}(\mathbf{A}).
\end{aligned}$$

If $\text{Range} d\mathbf{g}(\mathbf{A}) \neq \mathbb{R}^m$ then $d\mathbf{g}(\mathbf{A})$ has a non empty *left kernel* i.e.

$$\exists \langle \mathbf{v} | \ni \langle \boldsymbol{\lambda} | d\mathbf{g}(\mathbf{A}) = \mathbf{0} \iff \boldsymbol{\lambda} \in \ker \{d\mathbf{g}(\mathbf{A})^\dagger\}$$

and

$$\begin{aligned}
\text{Range} \{d\mathbf{g}(\mathbf{A})\}^\perp &= \{ \boldsymbol{\lambda} \mid \langle \boldsymbol{\lambda} | [d\mathbf{g}(\mathbf{A})] (\delta \mathbf{A}) \rangle = 0 \forall \delta \mathbf{A} \in \text{domain} \{d\mathbf{g}(\mathbf{A})\} \} \\
&= \{ \boldsymbol{\lambda} \mid \langle \boldsymbol{\lambda} | [d\mathbf{g}(\mathbf{A})] (\delta \mathbf{A}) \rangle = 0 \forall \delta \mathbf{A} \in \ker \{d\mathbf{g}(\mathbf{A})\}^c \}
\end{aligned}$$

If $\boldsymbol{\eta} \neq \mathbf{0} \in \ker \left\{ d\mathbf{g}(\mathbf{A})^\dagger \right\}$ then

$$\begin{aligned} \langle \boldsymbol{\eta} | [d\mathbf{g}(\mathbf{A})] (\delta \mathbf{A}) \rangle &= 0 \forall \delta \mathbf{A} \in V = \langle \boldsymbol{\eta} | \mathbf{w} \rangle = 0 \forall \mathbf{w} \in \text{Range} \{d\mathbf{g}(\mathbf{A})\} \\ &\Rightarrow \text{Range} \{d\mathbf{g}(\mathbf{A})\} \subset W \end{aligned}$$

thus $d\mathbf{g}(\mathbf{A})$ is not surjective. Further

$$\langle \boldsymbol{\lambda} | [d\mathbf{g}(\mathbf{A})] = \langle \boldsymbol{\lambda} + \boldsymbol{\eta} | [d\mathbf{g}(\mathbf{A})]$$

so functionals on $\text{Range} \{d\mathbf{g}(\mathbf{A})\}$ are not unique.

e.g. 1

$$\begin{aligned} g : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ g(x, y) &= x^2 + y^2 = c^2; \quad y = f(x) = +\sqrt{c^2 - x^2}; \quad [df(x)](\delta x) = \left[-x(c^2 - x^2)^{-\frac{1}{2}} \right](\delta x) \\ \langle \nabla g(x, y) | &= (2x, 2y) \\ \ker \{dg(x, y)\} &= \left\{ \alpha \begin{pmatrix} y \\ -x \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\} \cong \mathbb{R} \\ x(t) &= c \cos(\omega t + \omega_0), \quad y(t) = c \sin(\omega t + \omega_0) \\ \theta(t) &= \omega t + \omega_0 = \tan^{-1} \left\{ \frac{y(t)}{x(t)} \right\}; \omega \in \mathbb{R}, \quad \tan^{-1} \text{ is the principal value} \\ \text{for } x \neq 0 &\text{ then} \\ \theta(0) = \omega_0 &= \begin{cases} \tan^{-1} \left\{ \frac{y}{x} \right\}; x > 0 \\ \tan^{-1} \left\{ \frac{y}{x} \right\} - \pi; x < 0 \end{cases} \\ \text{for } x = 0 &\text{ then } \omega_0 = \begin{cases} \frac{\pi}{2}; y > 0 \\ -\frac{\pi}{2}; y < 0 \end{cases} \\ \mathcal{T}_{(x,y)}g &= \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \left\{ \omega \begin{pmatrix} -c \sin(\omega_0) \\ c \cos(\omega_0) \end{pmatrix} = \omega \begin{pmatrix} -y \\ x \end{pmatrix} \right. \\ &\quad \left. , \omega \in \mathbb{R} \equiv \alpha \begin{pmatrix} y \\ -x \end{pmatrix} \right\} \\ \mathcal{N}_{(x,y)}g &= [\mathcal{T}_{(x,y)}g]^\perp \cong \mathbb{R} \end{aligned}$$

e.g. 2

$$g : \mathbb{R}^3 \rightarrow \mathbb{R}; \quad g(x, y, z) = x^2 + y^2 + z^2 - c^2 = 0; \quad dg(x, y, z) : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$[dg(x, y, z)](\delta x, \delta y, \delta z) = (2x, 2y, 2z) \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} \equiv \langle \nabla g(x, y, z) | \delta \mathbf{x} \rangle$$

$$z = f(x, y) = +\sqrt{c^2 - x^2 - y^2} \quad f : \mathbb{R}^2 \rightarrow \mathbb{R}; \quad df(x, y) : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$[df(x, y)](\delta x, \delta y) = \frac{-1}{\sqrt{c^2 - x^2 - y^2}}(x, y) \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \equiv \langle \nabla f(x, y) | \delta \mathbf{x} \rangle$$

$$g(x, y, z) = F((x, y), f(x, y)) = 0 = x^2 + y^2 + z^2 - c^2 \equiv \text{graph} \{ f \}$$

$$\ker \{dg(x, y, z)\} = \left\{ \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} \middle| (2x, 2y, 2z) \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} = 0 \right\} \cong \mathbb{R}^2$$

$$\mathcal{N}_{(x,y,z)}g = \{ \alpha(x, y, z) | \alpha \in \mathbb{R} \} \cong \mathbb{R}$$

e.g. 3

$$g, dg(\mathbf{r}) : \mathbb{R}^3 \rightarrow \mathbb{R}^2 \equiv V \longrightarrow W; \quad dg(\mathbf{r})^\dagger : \mathbb{R}^2 \rightarrow \mathbb{R}^3 \equiv V^A \longrightarrow W^A$$

$$g_1(x, y, z) = x^2 + y^2 + z^2 = c_1^2$$

$$g_2(x, y, z) = x^2 + y^2 = c_2^2$$

$$\langle \nabla g_1(x, y, z) | = (2x, 2y, 2z)$$

$$\langle \nabla g_2(x, y, z) | = (2x, 2y, 0)$$

$$\ker \{d\mathbf{g}(x, y, z)\} = \left\{ \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} \middle| \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \subset V \equiv \mathbb{R}^3$$

$$\left. \begin{array}{l} 2x\delta x + 2y\delta y + 2z\delta z = 0 \\ 2x\delta x + 2y\delta y = 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} z\delta z = 0 \Rightarrow \delta z = 0 \text{ if } z \neq 0 \text{ and } \delta z \in \mathbb{R} \text{ if } z = 0 \\ \delta x = -\frac{y}{x}\delta y \text{ or } \delta y = -\frac{x}{y}\delta x \end{array} \right\}$$

$$\ker \{d\mathbf{g}(x, y, z)\} = \left\{ \begin{pmatrix} \delta x \\ -\frac{x}{y}\delta x \\ 0 \end{pmatrix} \middle| \begin{pmatrix} x & y & z \end{pmatrix} \neq \begin{pmatrix} 0 & 0 & 0 \end{pmatrix} \right\}$$

The kernel of the constraint

$$\ker \{\mathbf{A}\mathbf{g}(x, y, z)\} = \left\{ \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} \left| \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right. \right\} \subset V \equiv \mathbb{R}^3$$

$$\left. \begin{matrix} 2x\delta x + 2y\delta y + 2z\delta z = 0 \\ 2x\delta x + 2y\delta y = 0 \end{matrix} \right\} \Rightarrow \left. \begin{matrix} z\delta z = 0 \Rightarrow \delta z = 0 \text{ if } z \neq 0 \text{ and } \delta z \in \mathbb{R} \text{ if } z = 0 \\ \delta x = -\frac{y}{x}\delta y \text{ or } \delta y = -\frac{x}{y}\delta x \end{matrix} \right\}$$

$$x = 0, y \neq 0$$

$$2y\delta y + 2z\delta z = 0$$

$$2y\delta y = 0 \Rightarrow \delta y = 0 \Rightarrow 2z\delta z = 0$$

$$\ker \{d\mathbf{g}(x, y, z)\} = \left\{ \alpha \begin{pmatrix} 1 \\ -\frac{x}{y} \\ 0 \end{pmatrix} \left| \alpha \in \mathbb{R} \right. \right\}$$

$$\ker \{d\mathbf{g}(x, y, 0)\} = \left\{ \alpha \begin{pmatrix} 1 \\ -\frac{x}{y} \\ 0 \end{pmatrix}, \beta \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \left| \alpha, \beta \in \mathbb{R} \right. \right\}$$

$$\ker \{d\mathbf{g}(0, y, z)\} = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \left| \alpha \in \mathbb{R} \right. \right\}$$

$$\ker \{d\mathbf{g}(0, y, 0)\} = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \beta \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \left| \alpha, \beta \in \mathbb{R} \right. \right\}$$

$$\ker \{d\mathbf{g}(x, 0, z)\} = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \left| \alpha \in \mathbb{R} \right. \right\}$$

$$\ker \{d\mathbf{g}(x, 0, 0)\} = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \beta \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \left| \alpha, \beta \in \mathbb{R} \right. \right\}$$

$$\ker \{d\mathbf{g}(0,0,z)\} = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \beta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$$

$$\ker \{d\mathbf{g}(0,0,0)\} = \left\{ \alpha \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \beta \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha, \beta, \gamma \in \mathbb{R} \right\}$$

The Normal space $\mathcal{N}_{(x,y,z)} \mathbf{g}$: in order to see if the bases are lin. indep we consider

$$\begin{aligned} \alpha_1 \begin{pmatrix} 2x & 2y & 2z \end{pmatrix} + \alpha_2 \begin{pmatrix} 2x & 2y & 0 \end{pmatrix} &= 0 \\ \Rightarrow \alpha_1 + \alpha_2 &= 0 \text{ if } x \neq 0 \text{ or } y \neq 0 \text{ and } \alpha_1 z = 0 \\ \Rightarrow \alpha_1 = \alpha_2 &= 0 \text{ if } z \neq 0 \text{ so vectors are lin. indep.} \\ \Rightarrow \alpha_1 = -\alpha_2 &\text{ if } z = 0 \text{ so vectors are lin. dep.} \end{aligned}$$

$$\mathcal{N}_{(x,y,z)} \mathbf{g} = \text{linearspan} \{(x, y, z)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\} \subset W$$

$$\mathcal{N}_{(x,y,z)} \mathbf{g} = \text{linearspan} \{(x, y, z), (x, y, 0)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(x,y,0)} \mathbf{g} = \text{linearspan} \{(x, y, 0)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(0,y,z)} \mathbf{g} = \text{linearspan} \{(0, y, z), (0, y, 0)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(0,y,0)} \mathbf{g} = \text{linearspan} \{(0, y, 0)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(x,0,z)} \mathbf{g} = \text{linearspan} \{(x, 0, y), (x, 0, 0)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(x,0,0)} \mathbf{g} = \text{linearspan} \{(x, 0, 0)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(0,0,z)} \mathbf{g} = \text{linearspan} \{(0, 0, z)\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$\mathcal{N}_{(0,0,0)} \mathbf{g} = \{\phi\} = \text{range} \{d\mathbf{g}(\mathbf{r})\}$$

$$V^{\mathbf{A}} \equiv \mathbb{R}^3; W^{\mathbf{A}} \equiv \mathbb{R}^2$$

$$d\mathbf{g}(\mathbf{r})^\dagger : W^{\mathbf{A}} \rightarrow V^{\mathbf{A}}$$

Consider the Single Valued Decomposition (SVD)

$$\begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} \begin{pmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{pmatrix}^\dagger = \begin{pmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}^\dagger \begin{pmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \end{pmatrix} \begin{pmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{pmatrix}$$

$$\begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} = \alpha_1 \left| \begin{pmatrix} w_{11} & w_{12} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} v_{11}^{\mathbf{A}} & v_{12}^{\mathbf{A}} & v_{13}^{\mathbf{A}} \end{pmatrix} \right| + \alpha_2 \left| \begin{pmatrix} w_{21} & w_{22} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} v_{21}^{\mathbf{A}} & v_{22}^{\mathbf{A}} & v_{23}^{\mathbf{A}} \end{pmatrix} \right|$$

$$\begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix}^\dagger = \begin{pmatrix} 2x & 2x \\ 2y & 2y \\ 2z & 0 \end{pmatrix} \equiv dg(\mathbf{r})^\dagger$$

$$= \alpha_1 \left| \begin{pmatrix} w_{11} & w_{12} & w_{13} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} w_{11}^{\mathbf{A}} & w_{12}^{\mathbf{A}} \end{pmatrix} \right| + \alpha_2 \left| \begin{pmatrix} w_{21} & w_{22} & w_{23} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} w_{21}^{\mathbf{A}} & w_{22}^{\mathbf{A}} \end{pmatrix} \right|$$

$$dg(\mathbf{r}) dg(\mathbf{r})^\dagger = \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix}^\dagger = \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} \begin{pmatrix} 2x & 2x \\ 2y & 2y \\ 2z & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 4x^2 + 4y^2 + 4z^2 & 4x^2 + 4y^2 \\ 4x^2 + 4y^2 & 4x^2 + 4y^2 \end{pmatrix}$$

$$= \alpha_1^2 \left| \begin{pmatrix} w_{11} & w_{12} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} w_{11}^{\mathbf{A}} & w_{12}^{\mathbf{A}} \end{pmatrix} \right| + \alpha_2^2 \left| \begin{pmatrix} w_{21} & w_{22} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} w_{21}^{\mathbf{A}} & w_{22}^{\mathbf{A}} \end{pmatrix} \right|$$

$$dg(\mathbf{r})^\dagger dg(\mathbf{r}) = \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix}^\dagger \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix} = \begin{pmatrix} 2x & 2x \\ 2y & 2y \\ 2z & 0 \end{pmatrix} \begin{pmatrix} 2x & 2y & 2z \\ 2x & 2y & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 16x^2 & 16xy & 4xz \\ 16xy & 16y^2 & 4yz \\ 4xz & 4yz & 0 \end{pmatrix}$$

$$= \alpha_1^2 \left| \begin{pmatrix} v_{11} & v_{12} & v_{13} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} v_{11}^{\mathbf{A}} & v_{12}^{\mathbf{A}} & v_{13}^{\mathbf{A}} \end{pmatrix} \right| + \alpha_2^2 \left| \begin{pmatrix} v_{21} & v_{22} & v_{23} \end{pmatrix} \right\rangle \left\langle \begin{pmatrix} v_{21}^{\mathbf{A}} & v_{22}^{\mathbf{A}} & v_{23}^{\mathbf{A}} \end{pmatrix} \right|$$

$$\mathcal{N}_{(x,y,z)} \mathbf{g} \perp \mathcal{T}_{(x,y,z)} \mathbf{g}$$

$$\mathcal{T}_{(x,y,z)} \mathbf{g} = [\mathcal{N}_{(x,y,z)} \mathbf{g}]^\perp$$

$$\text{Rank} \{d\mathbf{g}(\mathbf{x})\} = \dim \{\mathcal{N}_{(x,y,z)} \mathbf{g}\}$$

e.g. 4

$$\begin{aligned}
g &: \mathbb{R}^3 \rightarrow \mathbb{R}^2 \\
g_1(x, y, z) &= x^2 + y^2 + z^2 = c_1^2 \\
g_2(x, y, z) &= x^2 - y^2 = c_2^2 \\
\langle \nabla g_1(x, y, z) | &= (2x, 2y, 2z) \\
\langle \nabla g_2(x, y, z) | &= (2x, -2y, 0) \\
\ker \{d\mathbf{g}(x, y, z)\} &= \left\{ \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} \middle| \begin{pmatrix} 2x & 2y & 2z \\ 2x & -2y & 0 \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \\
\mathcal{N}_{(x,y,z)}\mathbf{g} &= \text{linearspan} \{(2x, 2y, 2z), (2x, -2y, 0)\}
\end{aligned}$$

e.g.5

$$\begin{aligned}
g_1 &: \mathbb{R}^2 \rightarrow \mathbb{R} \\
g_1(x, y) &= (x^2 + y^2 - 1)^2 = 0 = x^4 + y^4 + 4x^2y^2 - 2x^2 - 2y^2 + 1 \\
\langle \nabla g_1(x, y) | &= 4 \left((x^2 + y^2 - 1)x, (x^2 + y^2 - 1)y \right) \\
\ker \{d\mathbf{g}_1[\mathbf{r}_0]\} &= \left\{ \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \middle| \begin{pmatrix} (x^2 + y^2 - 1)x & (x^2 + y^2 - 1)y \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \\
\mathcal{N}_{(x,y)}g_1 &= [\mathcal{T}_{(x,y)}g_1]^\perp
\end{aligned}$$

Let $(x, y) = (0, 1)$ then

$$\begin{pmatrix} (x^2 + y^2 - 1)x & (x^2 + y^2 - 1)y \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

which gives

$$\ker \{dg_1[(0, 1)]\} = \mathbb{R}^2.$$

A tangent vector $\begin{pmatrix} \delta x \\ \delta y \end{pmatrix}$ at the point $\mathbf{r}_0$ is defined by considering curves

$$\begin{aligned}
g_1(v(t)) &= 0, t \in [0, \delta] \\
v(0) &= \mathbf{r}_0
\end{aligned}$$

such curves could be $x(t) = \cos(\omega t + \omega_0)$; $y(t) = \sin(\omega t + \omega_0)$; $t \in [0, \frac{2\pi}{\omega})$, ω is an arbitrary constant and ω_0 determines the point where the tangent space is evaluated and must be a solution of

$$\begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} \frac{dv_x(t)}{dt} \\ \frac{dv_y(t)}{dt} \end{pmatrix}$$

This implies the necessary condition

$$\frac{dg_1(v(t))}{dt} = \left\langle \nabla g_1(x, y) \left| \frac{dv(t)}{dt} \right. \right\rangle = \nabla g_1(x, y)^t \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = 0$$

i.e. $\begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \in \ker \{dg_1[r_0]\}$. The curve g_1 is the unit circles with center at the origin, thus

its tangent at $(0, 1)$ is the line generated by $(1, 0)$ i.e the vectors $\left\{ \begin{pmatrix} \delta x \\ 0 \end{pmatrix} \right\}$, which form a subspace of $\ker \{dg_1[(0, 1)]\}$. This can be seen by considering the curve

$$x(t) = \cos(\omega t + \omega_0), \quad y(t) = \sin(\omega t + \omega_0),$$

where

$$x = x(0) = \cos(\omega_0) \Rightarrow \omega_0 = \cos^{-1}(x)$$

$$y = y(0) = \sin(\omega_0) \Rightarrow \omega_0 = \sin^{-1}(y)$$

which has derivatives

$$\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -\omega \sin(\omega t + \omega_0) \\ \omega \cos(\omega t + \omega_0) \end{pmatrix}$$

and thus tangent vectors at the point $(x, y) = (x(0), y(0))$

$$\left. \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} \right|_{t=0} = \begin{pmatrix} -\omega \sin(\omega_0) \\ \omega \cos(\omega_0) \end{pmatrix} = \omega \begin{pmatrix} -y \\ x \end{pmatrix}.$$

At the point $(0, 1) \equiv \omega_0 = \frac{\pi}{2}$ this gives

$$\left. \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} \right|_{t=0} = \begin{pmatrix} -\omega \\ 0 \end{pmatrix}; \omega \in \mathbb{R}$$

and thus

$$\mathcal{T}_{(0,1)g_1} = \text{linearspan} \left\{ \begin{pmatrix} 1 & 0 \end{pmatrix} \right\}.$$

$\mathbf{d}g_1 [(0, 1)] \equiv 0$ is not surjective and does not equal the normal space which is defined as

$$\mathcal{N}_{(x,y)}g_1 = [\mathcal{T}_{(x,y)}g_1]^\perp = \text{linearspan} \left\{ \begin{pmatrix} 0 & 1 \end{pmatrix} \right\}.$$

e.g.6

$$\begin{aligned} g_1 : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ g_1(x, y) &= x^2 - y^2 = 0 \\ \langle \nabla g_1(x, y) | &= 2(x, -y) \\ \ker \{ \mathbf{d}g_1[\mathbf{r}_0] \} &= \left\{ \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} \middle| \begin{pmatrix} x & -y \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\} \\ \mathcal{N}_{(x,y)}g_1 &= \text{linearspan} \left\{ \begin{pmatrix} x & -y \end{pmatrix} \right\} \end{aligned}$$

Let $(x, y) = (0, 0)$ then

$$\nabla g_1(0, 0) = \begin{pmatrix} x & -y \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

which also gives

$$\ker \{ \mathbf{d}g_1 [(0, 1)] \} = \mathbb{R}^2.$$

Now

$$\begin{aligned} \mathcal{T}_{(0,0)}g_1 &= \text{linearspan} \left\{ \begin{pmatrix} 0 & 1 \end{pmatrix} \right\} \\ \mathcal{N}_{(x,y)}g_1 &= [\mathcal{T}_{(0,0)}g_1]^\perp = \text{linearspan} \left\{ \begin{pmatrix} 1 & 0 \end{pmatrix} \right\} \end{aligned}$$

A point on the constraint manifold is called *normal* if $\{\nabla g_1(x, y)\}$ generates the normal space, which at the point $(0, 0)$ it does not, and *regular* if $\mathcal{T}_{(x,y)}g_1 = \ker \{ \mathbf{d}g_1 [(x, y)] \}$, which again at the point $(0, 0)$ is not true. *Normal implies regular.*

- $m = n$
- $m < n$
- $m > n$

I. BANACH SPACE

If a Banach space $\mathcal{B}$ has a basis $\{|e_j\rangle | 1 \leq j \leq \infty\}$ then

$$T : \mathcal{B} \rightarrow \mathbb{R}^\infty$$

where $\mathbb{R}^\infty$ is the space of sequences (not in general convergent) given by

$$|c\rangle = \sum_{1 \leq j \leq \infty} c_j |e_j\rangle \equiv |\mathbf{e}\rangle \mathbf{c}$$

this correspondence might not be 1-1. If the Banach space is a closure of a subset of distributions then

$$|c\rangle = \int c(x) |x\rangle \equiv |\mathbf{x}\rangle \mathbf{c}$$

II. EXTRA

As the example is quadratic it useful to consider the Hessian $d^2\mathbf{f}(A)$ and note that a sufficient condition that $\mathbf{A} + \delta\mathbf{A} \in \mathfrak{M}$ is that $\delta A \in \ker \{d\mathbf{g}(\mathbf{A})\}$ and

$$\left(\delta A | \widehat{\Phi^\dagger \Phi_\varphi} \delta A \right) = \mathbf{0}. \quad (46)$$

[

$$\left(\delta A | \Phi(\varphi_j)^\dagger \Phi(\varphi_j) \delta A \right) = \text{Tr} \left\{ \Phi(\varphi_j)^\dagger \Phi(\varphi_j) (\delta A)^2 \right\} = 0 \quad (47)$$

In order to satisfy this constraint a positive matrix to have a zero diagonal, which is not possible as the trace would then be zero

$$\rho = \text{Tr} \{ \Phi^\dagger \Phi_\varphi D_1 \} = \text{Tr} \{ \Phi^\dagger \Phi_\varphi D_2 \} \quad (48)$$

$$\Delta = D_1 - D_2 \quad (49)$$

$$\mathbf{0} = \text{Tr} \{ \Phi^\dagger \Phi_\varphi \Delta \} \quad (50)$$

]